

Great job, STEM explorer! Keep learning and reaching for the stars!



JULY 4th STEM CHALLENGE SHEET

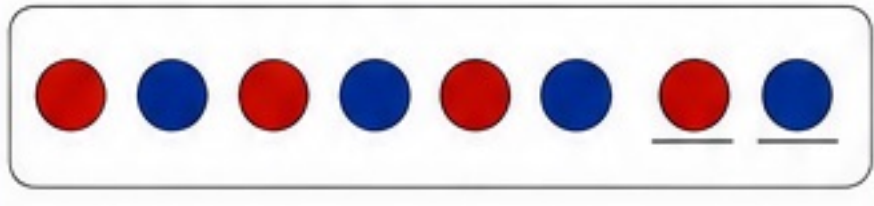
ANSWER KEY

1 Rocket Countdown 🚀
Help Nova count backwards.

10, 9, 8, 7, 6, 5, 4,
3, 2, 1, 0

(Answers in red)

2 Build a Firework Pattern 🎆
What comes next?



(Answers in red)

3 Count the Fireworks 🎆
How many fireworks do you see?



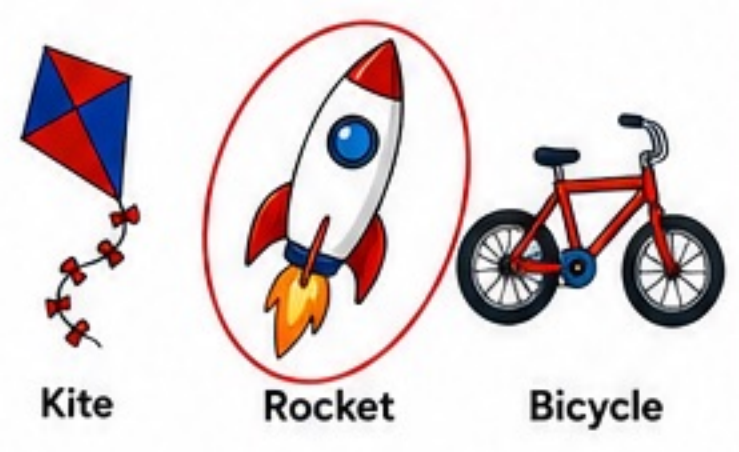
Answer: 7

How many stars?



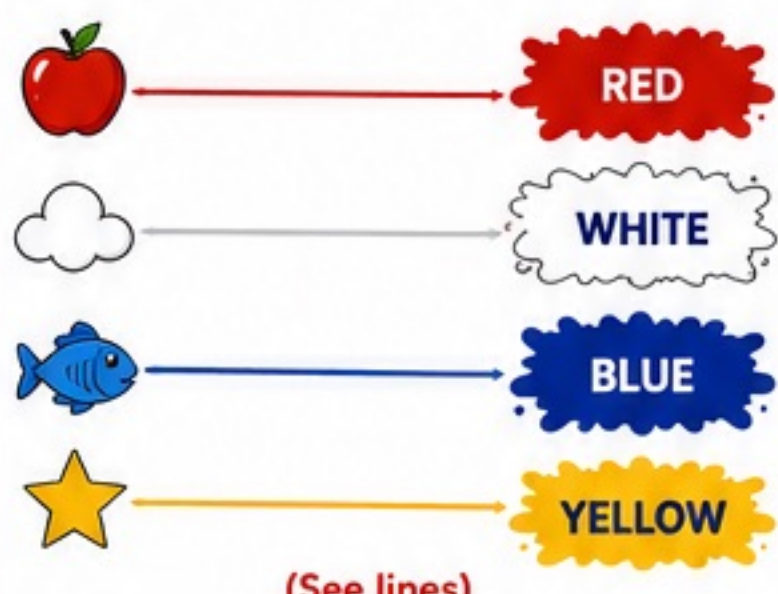
Answer: 8

4 Which Launches Higher? 🚀
Circle the object that can fly highest.



Answer: Rocket

5 Red, White & Blue Sort 🇺🇸
Draw a line to the correct color group.



(See lines)

6 STEM Observation Challenge 🔍
Look at the picture. Count.



Stars 10

Fireworks 4

Flags 1

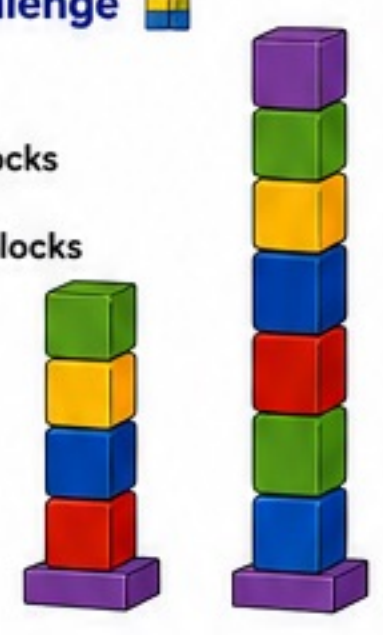
(Answers in red)

7 Build a Tower Challenge 🧱
Using blocks:

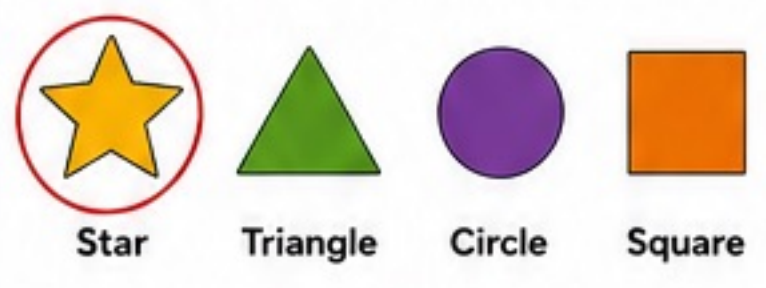
- ☒ Build a tower with 5 blocks
- ☒ Build a tower with 10 blocks

Which tower is taller?

The 10-block tower.



8 Firework Shapes 🎆
Circle the shape.



Which shape looks most like a firework?

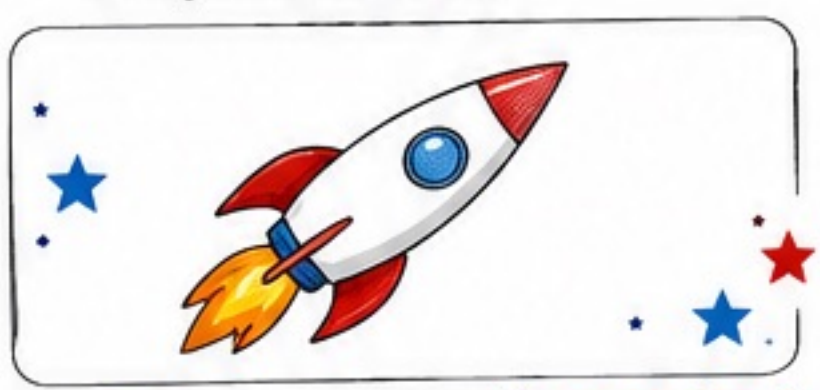
Answer: Star

9 Sink or Float Prediction 🧪
Predict before testing.

Item	Sink	Float
Cork	☐	☒
Coin	☒	☐
Pencil	☐	☒
Leaf	☐	☒

(Answers in red)

10 Nova's Mini Engineer Challenge 🧑🔬
Can you build a paper rocket?
Draw your rocket below.



(Answers will vary)

BONUS CHALLENGE

How many stars are on the American flag?



Answer: 100

SKILLS YOU'LL BUILD

- 13 Counting
- Observation
- Pattern Recognition
- Engineering Thinking
- Prediction Skills
- Early STEM Concepts
- Problem Solving

Great work, STEM Explorer! Keep shining! ★



NOVA'S STEM TIP

Scientists ask questions.
Engineers build solutions.
Today you can be both! ★

Name: _____ Date: _____

JULY 4th ENGINEERING & SCIENCE CHALLENGE SHEET

AGES 13-19 TEENS ANSWER KEY

Think. Solve. Innovate. Celebrate Freedom!

Great engineers solve problems. Great thinkers build the future. What will you create today?



1 FIREWORKS ENGINEERING
Design your own aerial firework shell. Fill in the specifications for your design.

Shell Diameter: 6 cm
Burst Shape: Chrysanthemum
Primary Color(s): Red, Blue
Secondary Color(s): Gold, White
Estimated Launch Height: 250 feet
Special Effect: Crackling stars

2 FIREWORKS MATH CHALLENGE
Solve the problems below.

a) $1,256 + 347 =$ 1,603
b) $2,480 - 965 =$ 1,515
c) $(325 \times 4) + 150 =$ 1,450
d) $3,600 \div 8 =$ 450
e) A firework kit has 192 rockets. You use $\frac{3}{8}$ of them. How many are left? 120

3 ENGINEERING DESIGN CHALLENGE
You have \$50 to design a safe and exciting firework show. List the materials you will use and the purpose of each.

Material	Cost	Purpose
Firework Shells	\$18	Create bursts of color
Fuses	\$5	Ignite shells safely
Launch Tube	\$10	Launch shells high
Safety Goggles	\$4	Protect eyes
Sparkler Pack	\$3	Handheld fun
Decorative Lights	\$5	Add visual effects

Total Cost: \$45

4 DECODE THE SECRET MESSAGE
Use the key to decode the message.

★ A ○ E 🇺🇸 I 🌟 O ❤️ U ! !

★ A ○ E 🇺🇸 I ★ A ○ E 🌟 O ! !

★ A ○ E 🇺🇸 I ★ A ○ E 🌟 O ! !

★ A ○ E ❤️ U 🌟 O ○ E 🇺🇸 I ★ A ! !

5 PATTERN LOGIC
What comes next in each sequence?

1) ● ● ● ● ● ● ● Red, Blue
2) ★ ★ ☆ ★ ☆ ☆ ★ ★, ☆
3) 🚀 ● 🚀 ● 🚀 ● 🚀, ●
4) 2, 6, 12, 20, 30, 42, 56
5) 1, 4, 9, 16, 25, 36, 49

6 DATA ANALYSIS
The table shows the number of fireworks launched in a show.

Type	Red	Blue	Gold	Total
Shells	24	18	12	<u>54</u>
Rockets	15	10	25	<u>50</u>
Sparklers	20	15	10	<u>45</u>
Total	<u>59</u>	<u>43</u>	<u>47</u>	<u>149</u>

a) Which type is used the most? Shells
b) How many gold fireworks in total? 47
c) If 20 more blue shells are added, what is the new total for blue? 43 + 20 = 63

7 STEM TRIVIA
Circle the best answer.

1. What gas gives fireworks their bright colors?
A) Oxygen B) Helium C) Copper D) Carbon Dioxide

2. Which law is used when a rocket is launched?
A) Newton's First Law B) Newton's Second Law C) Newton's Third Law D) Law of Gravity

3. What unit measures power of an engine?
A) Joule B) Newton C) Watt D) Liter

4. What is the main purpose of a fuse in fireworks?
A) To make sound B) To control the timing of the explosion
C) To change the color
D) To increase the size

8 MEASUREMENT CHALLENGE
A flagpole casts a shadow 18 feet long at the same time a person who is 5.5 feet tall casts a shadow 4 feet long. How tall is the flagpole?

Using similar triangles:
 $\frac{h}{18} = \frac{5.5}{4}$
 $h = (18 \times 5.5) / 4 = 99 / 4 = 24.75$ feet
Flagpole height = 24.75 feet

9 PARADE FLOAT BLUEPRINT
Design a parade float that represents freedom and innovation. Sketch your design below. Label the key parts and materials you would use.

Materials: Wood, Recycled Plastic, Paint, LED Lights, Solar Panel, Fabric Banners, Cardboard, Screws

10 INVENT INDEPENDENCE DAY 2050
Imagine the celebrations of the future!

Sample Ideas:

- Drone firework shows with eco-friendly lights
- Augmented reality (AR) parades
- Solar-powered celebration stages
- AI safety systems for large events
- Digital time capsules for future generations



“The best way to predict the future is to create it.”
- Peter Drucker



NOVA'S STEM TIP
Break big problems into small steps, test your ideas, and never stop improving!

SKILLS YOU'LL BUILD

Critical Thinking Problem Solving Math Skills Engineering Design Data Analysis Creativity & Innovation Communication & Planning

NOVA'S MISSION
Inspire young minds to explore, innovate, and build a brighter future!

JULY 4th ENGINEERING & SCIENCE CHALLENGE SHEET

ANSWER KEY

AGES 9-12

Think. Solve. Create. Celebrate Freedom!

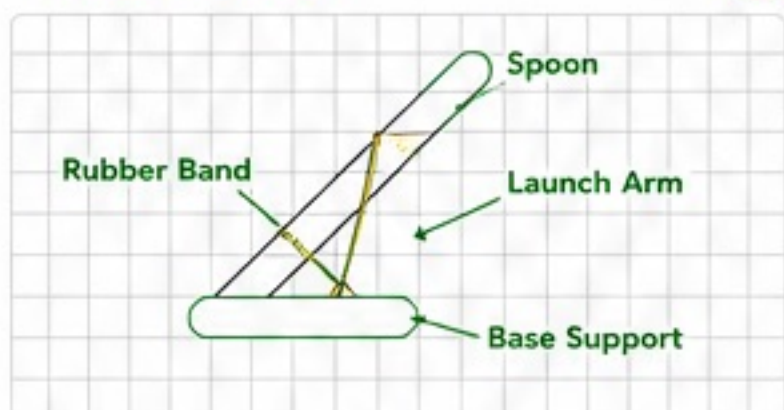
Great engineers solve problems.
Great thinkers build the future.
What will you create today?



1 BUILD A FIREWORK LAUNCHER

You have 10 craft sticks, 3 rubber bands, and a plastic spoon. Design a launcher that can launch a pom-pom as far as possible!

(Answers will vary)



How do you think your launcher works?

The rubber band stores energy when pulled back and releases it, pushing the pom-pom forward. ✓

2 FIREWORKS MATH

Solve the problems below.

- $489 + 276 = 765$ ✓
- $1,250 - 437 = 813$ ✓
- $(213 \times 5) + 125 = 1,190$ ✓
- $2,016 \div 8 = 252$ ✓
- A show uses 7 rows of 9 fireworks. How many fireworks are in the show?
63 ✓
- If each firework lasts 8 seconds, how long will the whole show last?
504 seconds (8 min 24 sec) ✓

3 ENGINEERING DESIGN CHALLENGE

Design a safe and eco-friendly firework that creates a beautiful effect.

Fill in the design planner below.

Shape of Firework: **Cone** ✓

Colors Used: **Red, White, Blue** ✓

Effect: **Starburst with glittering sparks** ✓

Materials (safe & eco-friendly):

Biodegradable paper casing, low-smoke powder, natural dyes ✓

How is it eco-friendly?

Uses recyclable materials and produces low smoke and less pollution. ✓

4 DECODE THE SECRET MESSAGE

Use the key to decode the message.

★	○	🇺🇸	☀️	❤️	!
A	E	I	O	U	!
★	○	🇺🇸	★	○	❤️
A	E	I	I	O	U
★	☆	🇺🇸	★	○	❤️
A	E	I	A	E	I
○	☀️	★	○	🇺🇸	★
A	E	E	E	E	A
★	○	❤️	★	🇺🇸	★
E	O	A	E	I	A

5 PATTERN LOGIC

What comes next?

- ■ ○ ● ■ ○ ● ■ ✓
- ★ ☆ ☆ ★ ☆ ☆ ☆ ☆ ☆ ☆ ✓
- 2, 4, 8, 16, 32, 64, 128 ✓
- 🇺🇸 ● 🇺🇸 ● 🇺🇸 ● 🇺🇸 ● ✓
- 1, 3, 6, 10, 15, 21, 28 ✓

6 DATA IN ACTION

Look at the chart and answer the questions.

Fireworks Sold at the Festival

Type	Friday	Saturday	Sunday	Total
Rockets	36	54	42	132
Fountains	28	32	25	85
Sparklers	45	60	50	155
Total	109	146	117	372

- Which firework type was sold the most on Saturday?
Sparklers ✓
- How many rockets were sold in total? **132** ✓
- How many more sparklers were sold than fountains on Sunday? **50 - 25 = 25** ✓

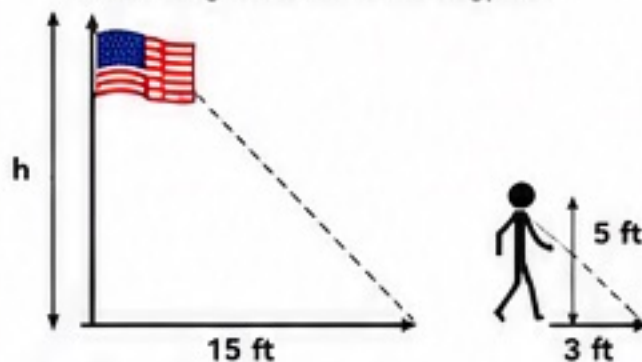
7 STEM TRIVIA

Circle the best answer.

- What gas helps fireworks create colors?
A) Oxygen B) Helium **C) Nitrogen** D) Carbon Dioxide
- Which law helps rockets move upward?
A) Law of Gravity B) Newton's First Law
C) Newton's Third Law D) Ohm's Law
- What unit measures force?
A) Watt B) Joule **C) Newton** D) Liter
- What is the main purpose of a fuse in fireworks?
A) To make sound
B) To control when it explodes
C) To change the color
D) To make it bigger

8 MEASUREMENT CHALLENGE

A flagpole casts a shadow 15 feet long at the same time a person who is 5 feet tall casts a shadow 3 feet long. How tall is the flagpole?



Using similar triangles:

$$\frac{h}{15} = \frac{5}{3} \rightarrow h = 15 \times \frac{5}{3} = 25$$

Flagpole height = **25 feet** ✓

9 SIMPLE CIRCUIT CHALLENGE

Draw a simple circuit to light up a LED (light-emitting diode). Use a battery, wires, and an LED.



What happens if the wires are not connected correctly?

The LED will not light up because the circuit is open. ✓

10 INVENT INDEPENDENCE DAY 2050

Imagine it's the year 2050! How will we celebrate Independence Day using technology?
Draw or describe your ideas below.

Sample Ideas:

- Drone light shows with eco-friendly fireworks
- Augmented reality (AR) parades
- Solar-powered celebration stages
- AI safety systems for large events
- Holographic historical reenactments ✓



SKILLS YOU'LL BUILD



Problem Solving



Math Skills



Engineering Design



Data Analysis



Creativity & Innovation



Critical Thinking

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Engineers solve problems, try new ideas, and never give up. You can be an engineer too!

